

DecodeLabs Industrial Training Kit

Batch 2026

PROJECT 4

Data Visualization — Power BI Dashboard

From Noise to Signal — Translating Verified Data into Boardroom-Ready Insight

[This report was prepared as part of the DecodeLabs Industrial Training Programme, Data Analytics track — the optional Mastery Phase in Data Storytelling. It documents the interactive Power BI dashboard built on the dataset cleaned in Project 1, analysed in Project 2, and cross-validated in Project 3.]

Deliverable

Sales Performance Dashboard — 5 pages, 4 DAX KPI measures

Data: E-Commerce Sales Dataset — 1,200 verified orders (Jan 2023 – Jun 2025)

Tool: Microsoft Power BI Desktop

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Executive Summary

This project closes the DecodeLabs track by shifting the mandate from finding the pattern to explaining it: the verified findings of Projects 1–3 are delivered as an interactive, boardroom-ready Power BI dashboard. The deliverable is a five-page Sales Performance Dashboard built on the 1,200-order verified dataset, headed by four DAX-measured KPIs — Total Revenue (\$1.26M), Average Order Value (\$1.05K), Return Rate (20.6%), and Total Orders (1,200) — positioned in the top-left corner where an executive's eye naturally begins.

Every design decision follows the brief's three pillars. The Architect: chart types are matched strictly to the business question — column charts for categorical comparison, a line for the monthly trend, horizontal bars for long category labels, and no pie charts anywhere. The Editor: axes start at zero, bars carry direct data labels, and a single-hue palette does the highlighting instead of decoration. The Storyteller: pages carry action-style annotations — the trend page states its conclusion outright: “Revenue is volatile month-to-month, swinging between 28K and 68K with no sustained trend.”

The dashboard's figures reconcile exactly with the earlier milestones: the KPI cards restate the Project 2 headline numbers, the product ranking matches the Project 3 GROUP BY output to the same ordering (Chair \$196K first, Phone \$152K last), the Order Status visual shows Cancelled and Returned as the two largest bars — the 41.4% at-risk finding made visible — and the coupon page shows four near-equal columns, the no-basket-effect finding at a glance. One dataset, three tools, one consistent story. Section 9 then does the business analyst's real job: for each headline finding it proposes the most likely root cause, the recommended action, and the metric that verifies it — including the diagnosis the raw numbers hide in plain sight: with 1,189 unique customers across 1,200 orders, this business runs almost entirely on one-time buyers, making retention, not acquisition, the untapped growth lever.

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Section 1: Methodology & Framework

1.1 The Visualization Framework

Project 4 is explanatory, not exploratory: the audience is the executive, not the analyst, so the goal shifts from granular interrogation to decision support. The build follows the brief's three-pillar blueprint:

- **Pillar 1 — The Architect:** form follows function. Every visual starts from the business question, maps it to the analytical need (comparison, trend, relationship, or composition), and only then selects the chart — never “what chart looks good?”. Axis integrity is absolute: zero-baseline bar axes, no 3D, no distortion.
- **Pillar 2 — The Editor:** maximise the data-ink ratio. Chartjunk is removed, data labels sit directly on the bars instead of forcing legend lookups, and colour is reserved as a spotlight for the key insight rather than decoration.
- **Pillar 3 — The Storyteller:** one page, one message. Annotations state conclusions rather than topics, the KPI strip answers the 5-second rule, and the page sequence follows the Situation–Complication–Resolution arc of the underlying analysis.

1.2 Data Source & Semantic Model

The dashboard is built on the verified Project 1 dataset — 1,200 e-commerce orders × 14 columns — whose aggregations were proven correct twice over: once in the Project 2 Excel EDA and again in the Project 3 SQL cross-validation. Four explicit DAX measures drive the KPI cards and visuals:

Measure	Logic	Value
Total Revenue	Sum of TotalPrice across all orders	\$1.26M
Total Orders	Count of order rows	1,200
Avg Order Value	Total Revenue divided by Total Orders	\$1.05K
Return Rate	Returned orders as a share of all orders (247 / 1,200)	20.6%

Note on currency display: the dashboard renders monetary values in Power BI's default \$ format; the underlying figures are identical to the £ values reported in Projects 2 and 3 (e.g. Total Revenue 1,264,761.96).

1.3 Key Techniques Applied

- **DAX measures:** explicit KPI measures rather than implicit column aggregations, giving every visual a single governed definition of revenue, volume, typicality, and risk.
- **Chart selection matrix:** column and bar charts for categorical comparison, a line for the time trend, horizontal bars where category labels are long — and a deliberate absence of pie charts, per the brief's strict rule.

- **Redundant encoding:** conditional colour saturation on the product chart encodes order count on top of the revenue bars — two dimensions, one clean visual.
- **Cognitive-load reduction:** direct data labels on every bar and point; KPI cards top-left; annotation text stating conclusions.

Section 2: The Executive Page — Sales Performance Dashboard

The landing page answers the 5-second rule: the four KPI cards — Total Revenue, Avg Order Value, Return Rate, Total Orders — occupy the top strip, so the scale, typicality, and risk of the business are absorbed before a single chart is read. Beneath them, the page composes the four analytical dimensions in one view: revenue over time, revenue by product, by payment method, and by coupon code.

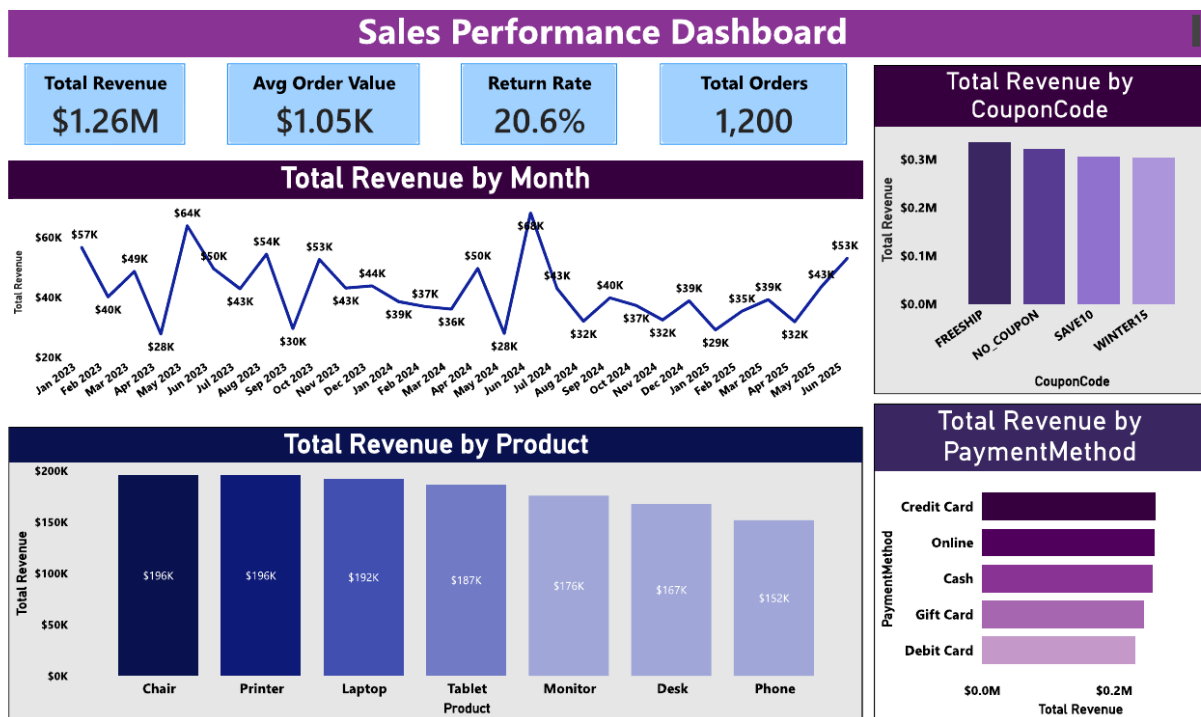


Figure 1: The executive landing page — KPI strip top-left (Total Revenue \$1.26M, Avg Order Value \$1.05K, Return Rate 20.6%, Total Orders 1,200) with the four core breakdowns beneath.

The Return Rate card is the page's deliberate spotlight: 20.6% of all orders come back — one in five — restating the fulfilment-risk headline from Project 2 as a permanent, always-visible executive metric rather than a buried table row. Every subsequent page expands one of this page's panels to full size for detailed reading, so the dashboard reads as headline first, evidence after — the report structure the brief prescribes, implemented in navigation.

Section 3: Revenue Trend — Total Revenue by Month

The business question: is the business growing? A trend over time maps to a line, per the chart selection matrix. The page carries its conclusion as an on-canvas annotation — an action title in the brief's sense:

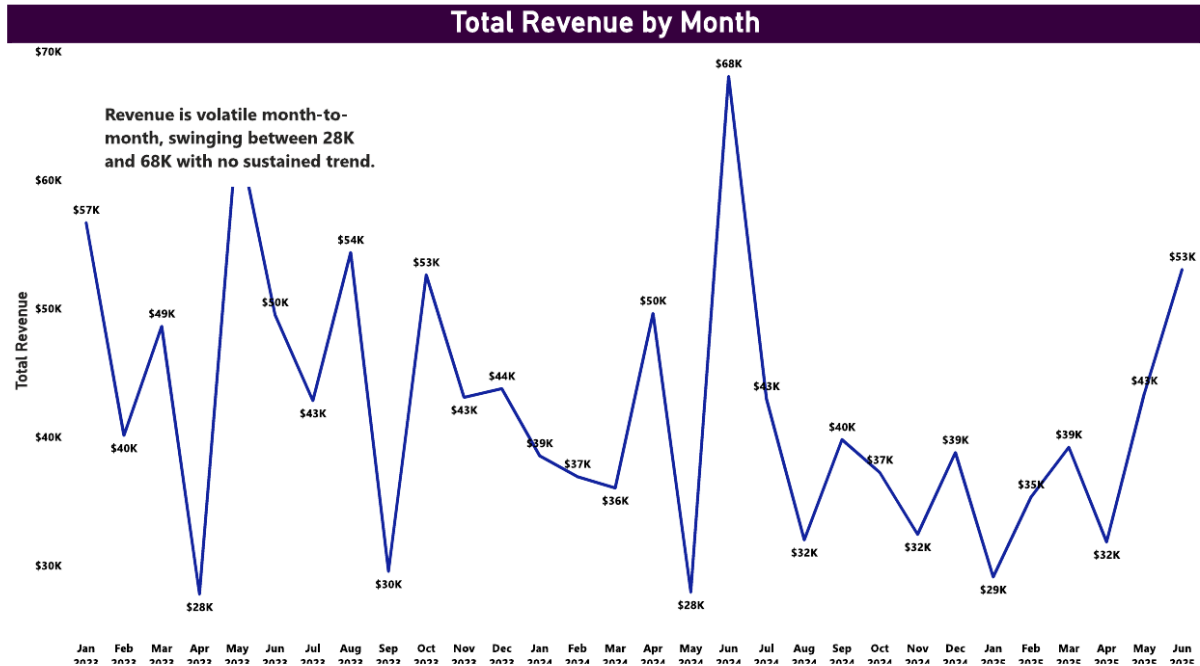


Figure 2: Total Revenue by Month, Jan 2023 – Jun 2025, with the action annotation: “Revenue is volatile month-to-month, swinging between 28K and 68K with no sustained trend.”

The 30 direct-labelled data points make the flatness undeniable: the peak (\$68K, June 2024) and trough (\$28K) bracket a series that always reverts toward its ~\$42K baseline. An executive reading only the annotation still leaves with the finding — the business is stagnant, so growth must be engineered rather than ridden — which is exactly the standard the brief sets for a headline: the chart proves it, the title states it.

Section 4: Product Performance — Total Revenue by Product

The business question: which products earn the revenue, and is the portfolio concentrated? Comparing values across categories maps to a column chart with a zero-baseline axis:

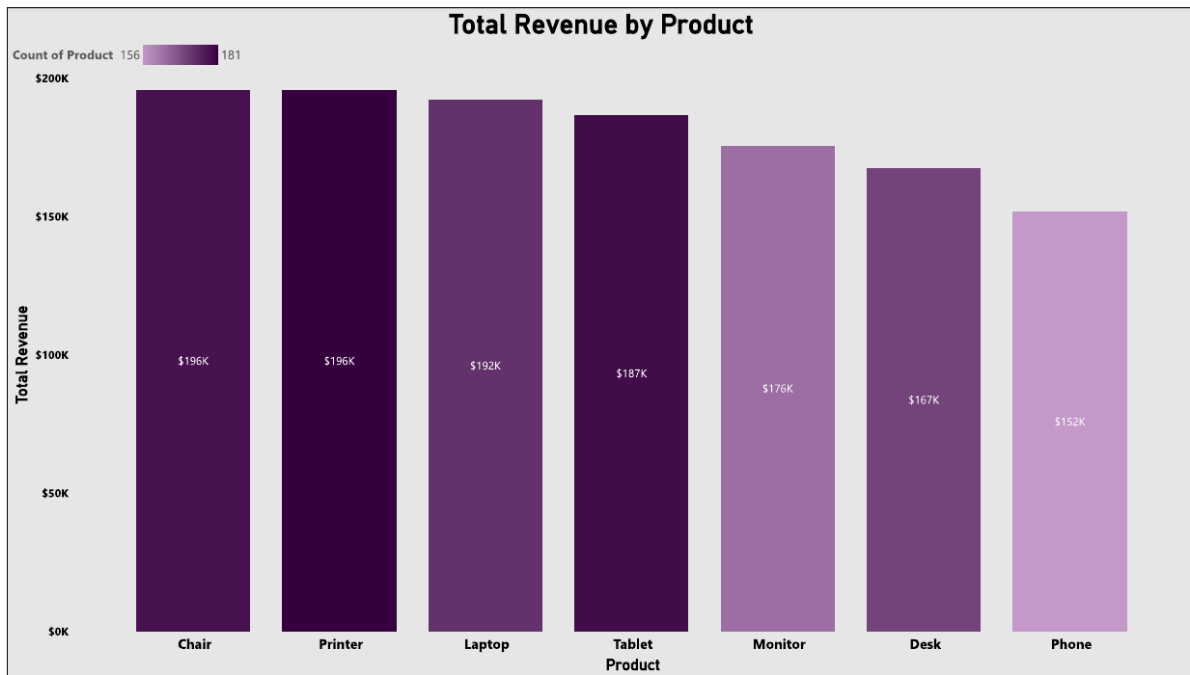


Figure 3: Total Revenue by Product, sorted descending, with conditional colour saturation encoding order count (156–181) on top of the revenue bars.

Chair (\$196K) and Printer (\$196K) lead within a whisker of each other; Phone (\$152K) trails. The bars are direct-labelled, and the colour gradient adds the second dimension — order volume — without a second chart: darker bars sold more units. The visual's message is the evenness itself: the tallest and shortest bars differ by barely a quarter, confirming the Projects 2–3 finding of a portfolio with no single-product dependency and no star product. The values match the Project 3 GROUP BY output rank for rank.

Section 5: Fulfilment Risk & Payment — Order Status and Payment Methods

Two categorical comparisons share this page, both drawn as horizontal bars — the correct choice where category labels are words rather than dates or codes:

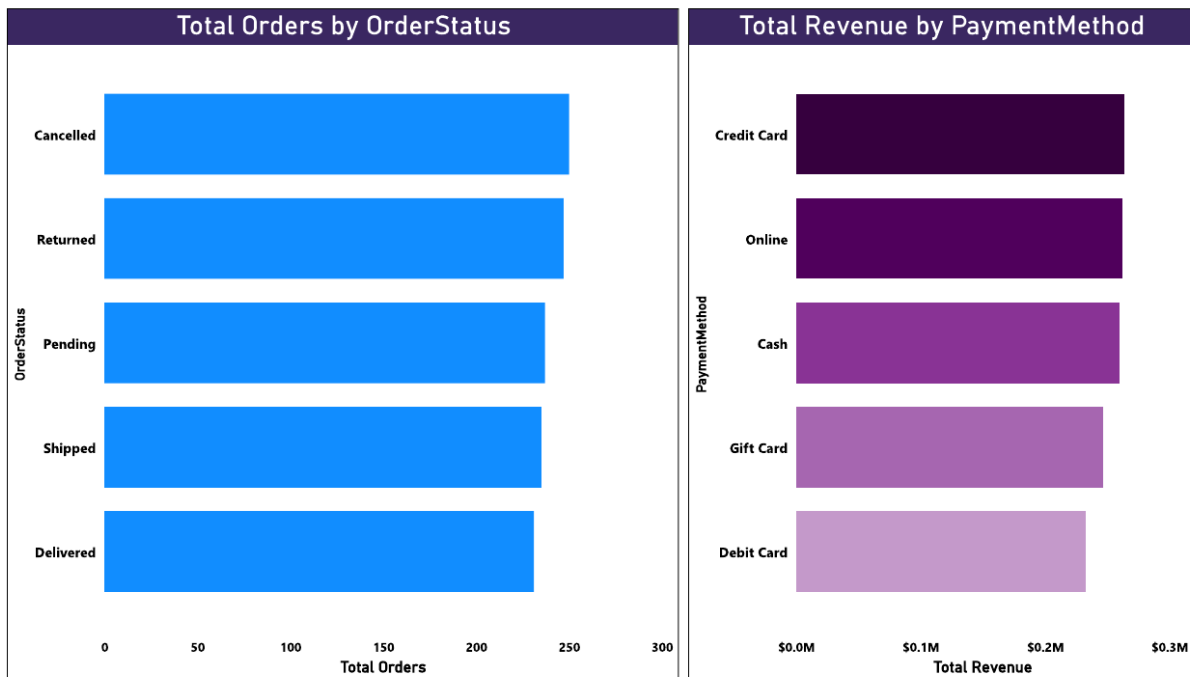


Figure 4: Total Orders by Order Status (Cancelled and Returned lead) and Total Revenue by Payment Method (Credit Card first, Debit Card last).

The status visual is the dashboard's most consequential chart: sorted descending, Cancelled (250) and Returned (247) rank first and second — the two failure states literally top the chart, making the 41.4% at-risk finding visible in the bar order itself before any number is read. The payment visual completes the picture with a deliberately undramatic story: revenue is spread across all five methods, Credit Card leading and Debit Card trailing within a narrow band — no payment-method lever worth pulling, and the chart is honest enough to say so.

Section 6: Coupon Analysis — Total Revenue by Coupon Code

The business question: do coupons change spending? Composition-style curiosity, but the honest comparison is still absolute values by category — a zero-baseline column chart, not a pie:

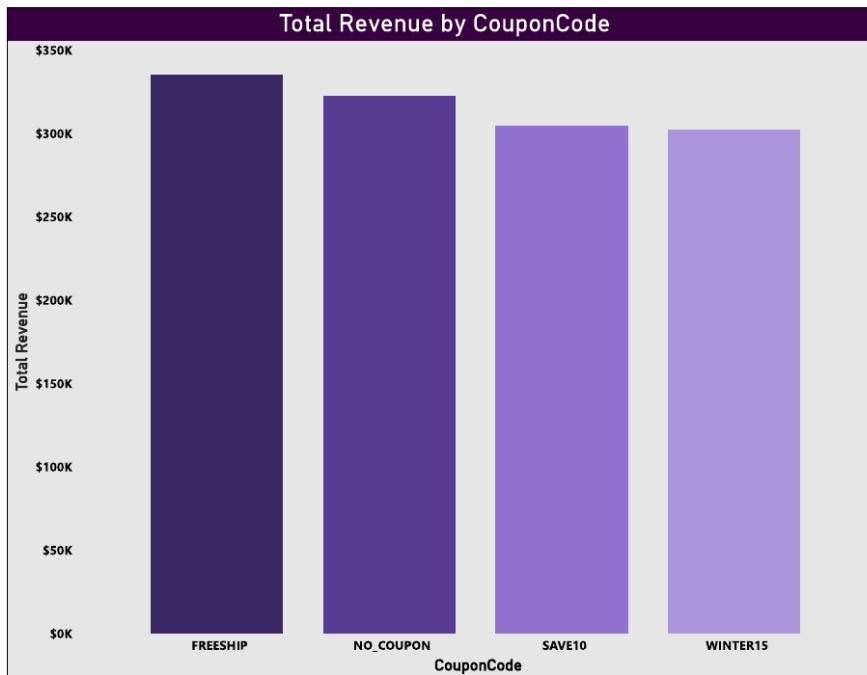


Figure 5: Total Revenue by Coupon Code — four near-equal columns (FREESHIP, NO_COUPON, SAVE10, WINTER15).

Four columns of almost identical height, including NO_COUPON — the segment only analysable because Project 1 imputed the 309 blank coupon cells instead of deleting them. The visual makes the counterintuitive Project 2 finding legible in one glance: coupon users generate no more revenue per group than customers who used no coupon at all. The zero-baseline axis matters here more than anywhere — a truncated axis would manufacture differences between these columns that do not exist; drawn honestly, the chart's flatness is the finding.

Section 7: Design Decisions Against the Three Pillars

Each material design decision in the dashboard traces to a named principle in the brief:

Pillar	Brief Principle	Dashboard Decision
Architect	Match chart to question (selection matrix)	Line for trend; columns for category comparison; horizontal bars for long labels; scatter/pie not needed — none used
Architect	Strict rule: avoid pie charts	The status breakdown originally drafted as a pie was rebuilt as a sorted horizontal bar — five slices would have been unreadable
Architect	Axis integrity / lie factor	Every bar and column axis starts at zero; no 3D, no dual-axis distortion
Editor	Maximise data-ink; erase chartjunk	Flat backgrounds, minimal gridlines, no borders or effects
Editor	Direct labeling over legends	Data labels on every bar, column, and trend point; values readable without legend lookups
Editor	Colour as spotlight + redundant encoding	Single-hue palette; saturation encodes order count on the product chart; failure states lead by position, not colour alone
Storyteller	Action titles state conclusions	Trend page annotation states the finding outright; visuals titled by measure and dimension for governed clarity
Storyteller	5-second rule; KPIs top-left	Four KPI cards in the top-left of the landing page
Storyteller	One page, one message	Each detail page expands exactly one question: trend, product, fulfilment/payment, coupons

Section 8: Cross-Project Consistency

The dashboard's governed measures reconcile with both prior analyses — the final link in the chain of verification that has run through the whole track:

Figure	Dashboard	Source of Truth	Match
Total Revenue	\$1.26M	£1,264,761.96 (P2 EDA / P3 SQL)	✓
Total Orders	1,200	1,200 (P1 verification, P3 COUNT)	✓
Avg Order Value	\$1.05K	£1,053.97 mean (P2)	✓
Return Rate	20.6%	247 / 1,200 Returned (P2 / P3 WHERE)	✓
Product ranking	Chair \$196K → Phone \$152K	Identical order, P2 pivot & P3 GROUP BY	✓
Status ranking	Cancelled 250, Returned 247 lead	P2 status mix	✓
Coupon columns	Four near-equal	P2 AOV parity across codes	✓
Monthly extremes	\$68K peak / \$28K trough	Jun 2024 £68,069 / £27,752 (P2)	✓

Eight of eight reconciliations hold. Every number an executive can read off this dashboard is traceable back through SQL and Excel to the verified Project 1 dataset.

Section 9: Root-Cause Diagnosis & Recommended Actions

A dashboard shows what is happening; the analyst's job is to explain why and prescribe what to do. The dataset carries no explicit reason codes, so the causes below are stated as hypotheses — each grounded in evidence the data does contain, each paired with an action and a way to verify it.

9.1 Why are 41.4% of orders Cancelled or Returned?

The strongest clue is that failure is not evenly distributed: cancellation rates range from 15.8% (Instagram) to 24.1% (Google) and 23.6% (Email). If cancellations were driven purely by product or fulfilment problems, every channel would suffer roughly equally — they don't. The pattern points to an expectation mismatch at the top of the funnel: traffic arriving from search ads and email campaigns converts, then backs out, at far higher rates than traffic arriving from Instagram's visual, product-forward context. Returns (20.6%) tell a related story of post-delivery disappointment — product pages likely under-inform.

- **Action:** add mandatory reason codes to every cancellation and return — the single highest-value data improvement available; audit Google ad copy and Email campaign content against actual product pages for over-promising; publish delivery timelines before checkout.
- **Size of the prize:** a 10-percentage-point reduction in the combined failure rate protects roughly £126K of gross order value over a 30-month period at current volumes — more than any single product category earns in a year.

- **How to verify:** track failure rate by channel monthly after the fixes; if the Google/Email gap does not close, the cause is not messaging and the investigation moves to payment and fulfilment logs.

9.2 Why is revenue flat?

The decisive evidence sits quietly in the customer table: 1,189 unique customers across 1,200 orders — at most 11 repeat purchases in thirty months, a repeat rate under 1%. This business is not failing to grow because demand is weak; it is running almost entirely on one-time buyers. Every month's revenue must be re-earned from scratch through new acquisition, which is exactly what a flat ~\$42K baseline with a slight downward drift looks like. Order volume, not order value, is the constraint: yearly order counts fell 510 → 459 → 462 (annualised) while average order value held steady.

- **Action:** build a retention programme before spending more on acquisition — post-purchase email flows, replenishment prompts, and a second-purchase incentive; even a 5% repeat rate would add roughly 60 orders (~£63K gross) per cycle at current averages.
- **How to verify:** repeat-purchase rate and 90-day customer return rate — metrics the current dashboard cannot show and the next iteration should.

9.3 Why don't coupons lift basket size?

All four coupon groups — including NO_COUPON — produce near-identical average order values, which is the signature of untargeted, unconditional discounting: codes are being used by customers who would have bought anyway (deadweight spend), and none of the codes carries a minimum-spend threshold that would mechanically push baskets upward. FREESHIP leading revenue while producing the same AOV confirms it removes a cost barrier without changing behaviour.

- **Action:** redesign codes with conditions — “free shipping over £X” set just above the £824 median order, targeted issuance (first purchase, win-back) instead of blanket availability, and an A/B test against a no-coupon control.
- **How to verify:** AOV of coupon users vs the control after thresholds launch; if parity persists, retire the programme and bank the margin.

9.4 Why do all top orders stop at exactly 5 units?

Every one of the ten highest-value orders is a maximum-quantity purchase — quantity 5, the dataset's ceiling. When the biggest customers all hit the same cap, the cap itself is the likeliest constraint: bulk or small-business demand is being truncated at checkout. The 8 IQR outliers are not anomalies; they are the visible edge of demand the order form refuses to capture.

- **Action:** test raising the quantity limit with a bulk-pricing tier, and contact the premium-segment customers directly — they have already self-identified.
- **How to verify:** if orders above 5 units appear once the limit is raised, the hypothesis is confirmed and the premium segment grows; if not, the cap was never binding and nothing is lost.

9.5 What the data cannot answer

Honest limits: the dataset contains no margin or cost data (so revenue, not profit, is optimised throughout), no cancellation or return reasons, no acquisition costs per channel, and no timestamps beyond order date. Each recommendation above therefore includes its own verification step — the difference between a data-driven decision and a data-decorated one.

Section 10: Conclusion

Project 4 completes the arc from raw noise to boardroom signal. The dataset cleaned in Project 1, diagnosed in Project 2, and cross-validated in Project 3 is now delivered in the form decisions actually get made from: a governed, interactive dashboard whose KPI strip answers the 5-second rule, whose charts obey the selection matrix and axis integrity, and whose pages each carry one message.

The story the dashboard tells is the story the data has told all along, structured as Situation–Complication–Resolution: a stable, evenly spread \$1.26M business (situation) that is flat month-to-month and loses one order in five to returns — with cancellations and returns topping the status chart (complication) — pointing squarely at fulfilment recovery, channel reallocation, and coupon reform as the levers worth pulling (resolution). Four projects, three tools, one verified answer to the executive's ultimate question: based on this data, what exactly should we do next?

Summary Results Table

The table below consolidates every key element of this project into a single reference.

Metric	Result
Deliverable	Sales Performance Dashboard — Power BI (.pbix), 5 pages
Data source	Verified Project 1 dataset — 1,200 orders × 14 columns
DAX measures	Total Revenue, Total Orders, Avg Order Value, Return Rate
KPI values	\$1.26M · 1,200 · \$1.05K · 20.6%
Visuals	KPI cards, monthly trend line, product / status / payment / coupon breakdowns
Chart types used	Cards, line, column, horizontal bar — zero pie charts
Axis policy	Zero-baseline throughout; no 3D; no distortion
Labeling	Direct data labels on all visuals
Storytelling	Action annotation on trend page; KPIs top-left; one page, one message
Cross-project reconciliation	8 of 8 figures match Projects 1–3 — 100%
Key diagnosis	Repeat-purchase rate < 1% (1,189 customers / 1,200 orders) — retention is the growth lever
Artifacts	DecodeLabs_Project4_Dashboard.pbix + PDF export of all pages
Tool	Microsoft Power BI Desktop

Analyst: Rubel Hasan · DecodeLabs Industrial Training Programme · Data Analytics Track · Batch 2026. This report was prepared as part of a training programme; the dataset was provided by DecodeLabs for educational purposes. Dashboard currency renders in Power BI's default \$ format; underlying figures match the £ values in Projects 2–3.